Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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7 Proving Non-Conditional Statements

Prove the following statements.

2. Suppose $x \in \mathbb{Z}$. Then x is odd if and only if $3x + 6$ is odd.

Proof. First we prove that if x is odd, then $3x + 6$ is odd. Suppose x is odd. Then, by definition of an odd number, $x = 2k+1$ for some integer k. Thus $3x+6 = 3(2k+1)+6 = 6k + 9 = 2(3k + 4) + 1$. Therefore $3x + 6$ is odd by definition of an odd number.

Conversely, we need to prove that if $3x + 6$ is odd, then x is odd. We use contrapositive proof. Suppose x is not odd. Then x is even, so $x = 2k$ for some integer k. Thus $3x + 6 = 3(2k) + 6 = 6k + 6 = 2(3k + 3)$. Therefore $3x + 6$ is even by definition of an even number. Thus $3x + 6$ is not odd. □

4. Given an integer a, then $a^3 + a^2 + a$ is even if and only if a is even.

Proof. First we prove that if $a^3 + a^2 + a$ is even, then a is even. We use contrapositive proof. Suppose a is not even. Then a is odd. Thus $a = 2k + 1$ for some integer k. Hence, $a^3 + a^2 + a = (2k + 1)^3 + (2k + 1)^2 + (2k + 1)$. After expanding we get $8k^3 + 16k^2 + 12k + 3 = 2(4k^3 + 8k^2 + 6k + 1) + 1$. Therefore, $a^3 + a^2 + a$ is odd by definition of an odd number.

Conversely, we need to prove that if a is even, then $a^3 + a^2 + a$ is even. Suppose a is even. Then $a = 2k$ for some integer k. Then $a^3 + a^2 + a = (2k)^3 + (2k)^2 + (2k) = 8k^3 + 4k^2 + 2k = 2(4k^3 + 2k^2 + k)$. Therefore, $a^3 + a^2 + a$ is even by definition of an even number. □

6. Suppose $x, y \in \mathbb{R}$. Then $x^3 + x^2y = y^2 + xy$ if and only if $y = x^2$ or $y = -x$.

Proof. First we prove that if $x^3 + x^2y = y^2 + xy$ then $y = x^2$ or $y = -x$.
Assume $x^3 + x^2y = y^2 + xy$. From this we get $x^2(x+y) = y(x+y)$. Subtracting $y(x+y)$ from both sides, we get $x^2(x+y) - y(x+y) = 0$. Factoring gives us $(x+y)(x^2-y) = 0$. Then either $(x+y) = 0$ or $(x^2 - y) = 0$. Therefore $y = -x$ or $y = x^2$.

Conversely, we need to prove that if $y = x^2$ or $y = -x$ then $x^3 + x^2y = y^2 + xy$, we look at both cases.

Case 1. Assume $y = x^2$. By substituting this value for y in the equation we get $x^3 + x^4 = x^4 + x^3$. So both sides equal $x^3 + x^4$, hence the equation holds.

Case 2. Assume $y = -x$. By substituting this value for y in the equation we get $x^3 - x^3 = x^2 - x^2$. So both sides equal 0, hence the equation holds.

Therefore $x^3 + x^2y = y^2 + xy$ both when $y = x^2$ and $y = -x$. □

8. Suppose $a, b \in \mathbb{Z}$. Prove that $a \equiv b \pmod{10}$ if and only if $a \equiv b \pmod 2$ and $a \equiv b \pmod 5$.

Proof. First we prove that if $a \equiv b \pmod{10}$ then $a \equiv b \pmod 2$ and $a \equiv b \pmod 5$. Assume $a \equiv b \pmod{10}$. Then $(a-b) = 10k$ for some integer k. Hence $(a-b) = 2(5k)$ and $(a-b) = 5(2k)$. In other words, $a \equiv b \pmod 2$ and $a \equiv b \pmod 5$.

Conversely, we need to prove that if $a \equiv b \pmod 2$ and $a \equiv b \pmod 5$ then $a \equiv b \pmod{10}$.
Assume $a \equiv b \pmod 2$ and $a \equiv b \pmod 5$. Then $(a-b) = 2m$ and $(a-b) = 5n$ for integers m and n. Then $2m = 5n$. Hence $5n$ is even by definition. Since 5 is odd, if n were odd then $5n$ would be odd; thus n is even, that is, $n = 2k$ for some integer k. So we have $a - b = 5n = 5(2k) = 10k$. Therefore $a \equiv b \pmod{10}$ by definition of congruence. □

10. If $a \in \mathbb{Z}$, then $a^3 \equiv a \pmod 3$.

Proof. Assume $a \in \mathbb{Z}$. Notice that $(a^3 - a) = a(a-1)(a+1)$. By the division algorithm, a must be one of the forms $a = 3k$, $a = 3k+1$, or $a = 3k+2$ for some integer k. Now consider each case.

Case 1. Assume $a = 3k$. Then $3 \mid a$, so $3 \mid a(a-1)(a+1)$.
Case 2. Assume $a = 3k+1$. Then $a - 1 = 3k$, so $3 \mid a(a-1)(a+1)$.
Case 3. Assume $a = 3k+2$. Then $a + 1 = 3k + 3 = 3(k+1)$. Thus $3 \mid a(a-1)(a+1)$.

In all three cases $3 \mid a(a-1)(a+1)$. Since $a(a-1)(a+1) = a^3 - a$, we conclude that $a^3 \equiv a \pmod 3$. □

12. There exist a positive real number x for which $x^2 < \sqrt{x}$.

Proof. Take $x = \frac{1}{2}$. Substituting this value for x, we get $\frac{1}{4} < \frac{1}{\sqrt{2}}$. Since $\frac{1}{2} \in \mathbb{R}^+$, we proved there exists a positive real number x for which $x^2 < \sqrt{x}$. $\square$

14. Suppose $a \in \mathbb{Z}$. Then $a^2 \mid a$ if and only if $a \in \{-1, 0, 1\}$.

Proof. First we prove that if $a^2 \mid a$ then $a \in \{-1, 0, 1\}$.
Suppose $a^2 \mid a$. Then $a = a^2k$ for some integer k. Subtracting a^2k from both sides we get $a - a^2k = 0$. Factoring the left sides give us $a(1 - ak) = 0$. Then either $a = 0$ or $1 - ak = 0$. Consider $1 - ak = 0$. Adding ak to both sides we get $1 = ak$. Thus $a \mid 1$. But 1 has one positive divisor which is 1, and a negative divisor which is -1. Therefore $a = 1$, $a = -1$, or $a = 0$, that is $a \in \{-1, 0, 1\}$.

Conversely, suppose $a \in \{-1, 0, 1\}$. Now we look into the three cases individually.

Case 1. Suppose $a = 1$. Then $a^2 = 1$, and since $1 = 1 \cdot 1$, we have $1 \mid 1$, that is, $a^2 \mid a$.
Case 2. Suppose $a = -1$. Then $a^2 = 1$, and since $-1 = 1 \cdot (-1)$, we have $1 \mid -1$, that is, $a^2 \mid a$.
Case 3. Suppose $a = 0$. Then $a^2 = 0$, and since $0 = 0 \cdot 0$, we have $0 \mid 0$ by definition of divisibility, that is, $a^2 \mid a$.

The three cases above show that if $a \in \{-1, 0, 1\}$ then $a^2 \mid a$. $\square$

16. Suppose $a, b \in \mathbb{Z}$. If ab is odd, then $a^2 + b^2$ is even.

Proof. We prove the contrapositive. Suppose $a^2 + b^2$ is not even. Then $a^2 + b^2$ is odd. Since exactly one of a^2,b^2 is odd, without loss of generality suppose a^2 is odd and b^2 is even. Hence a is odd and b is even. Thus $a = 2k+1$ and $b = 2l$ for integers k and l. So it follows that $ab = (2k+1)(2l) = 4kl + 2l = 2(2kl + l)$. Therefore ab is even by definition of an even number. $\square$

18. There is a set X for for which $\mathbb{N}$ and $\mathbb{N} \subseteq X$.

Proof. Take $X = \mathbb{N} \cup \{\mathbb{N}\}$. Since X contains every element of $\mathbb{N}$, we have $\mathbb{N} \subseteq X$. Since X explicitly includes $\{\mathbb{N}\}$, we have $\mathbb{N} \in X$. $\square$

20. There exists an $n \in \mathbb{N}$ for which $11 \mid (2^n - 1)$.

Proof. Take $n = 10$. Then $2^n - 1 = 1023 = 11 \cdot 93$. Therefore $11 \mid (2^n - 1)$ by definition of divisibility. $\square$